



**JMEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY,
JAMSHORO**



**DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 11**

To observe 741 Op-amp as a Comparator (Zero level detector)

Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

Performance:

Objective: To be familiar with the operation of the Op-amp as a Comparator. .

Apparatus:

- i. Op-amp 741 IC
- ii. Resistors
- iii. Oscilloscope
- iv. DC voltage Power Supply

Theoretical Background:

A voltage comparator is electronic circuits that compares two input voltages and lets you know which of the two is greater. It's easy to create a voltage comparator from an op amp, because the polarity of the op-amp's output circuit depends on the polarity of the difference between the two input voltages. A comparator is a device with two input terminals, inverting and non- inverting, and an output that usually swings from rail to rail. So is an op amp. A comparator has low offset, high gain, and high common-mode rejection. So does an op amp.

CIRCUIT DIAGRAM:

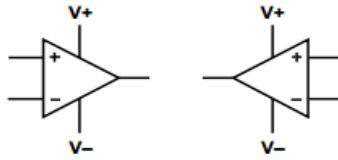


Figure-6.1 OP-Amp symbol

THEORY:

In the voltage-comparator circuit, first a reference voltage is applied to the inverting input (V_-); then the voltage to be compared with the reference voltage is applied to the no inverting input. The output voltage depends on the value of the input voltage relative to the reference voltage, as follows:

Input Voltage	Output Voltage
Less than reference voltage	Negative
Equal to reference voltage	Zero
Greater than reference voltage	Positive

Note that the voltage level for both the positive and negative output voltages will be about 1 V less than the power supply. Thus, if the op-amp power supply is 9 V, the output voltage will be +8 V if the input voltage is greater than the reference voltage, 0 V if the input voltage is equal to the reference voltage and -8 V if the input voltage is less than the reference voltage.

COMPARATOR APPLICATIONS

1. ZERO LEVEL DETECTION:

One application of an op-amp used as a comparator to determine when an input voltage exceeds a certain level. Notice that the inverting (-) input is grounded to produce a zero level and that the input signal voltage is applied to the non-inverting (+) input. Because of the high open-loop voltage gain, a very small difference voltage between the two inputs drives the amplifier into saturation, causing the output voltage to go its limit.

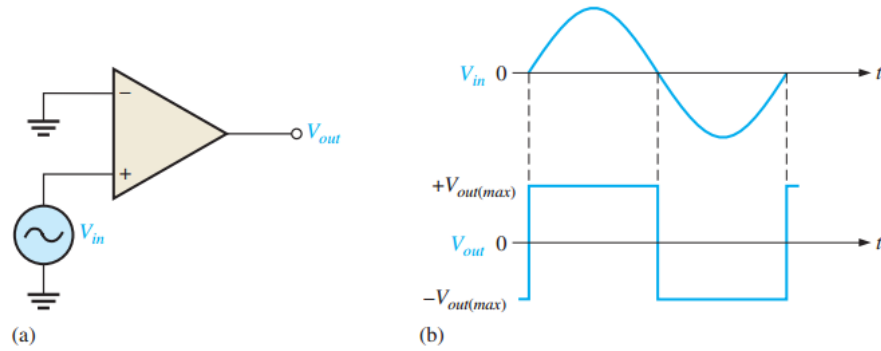


Figure-6.2 Zero level detector circuit and input output waveform

When the sine wave is negative, the output is at its maximum negative level. When the sine wave crosses “0”, the amplifier is driven to its opposite state and the output goes to its maximum positive level. The zero level detectors can be used as a squaring circuit to produce a square wave from a sine wave.

2. NON-ZERO LEVEL DETECTION:

When the sine wave is negative, the output is at its maximum negative level. When the sine wave crosses “0”, the amplifier is driven to its opposite state and the output goes to its maximum positive level. The zero level detectors can be used as a squaring circuit to produce a square wave from a sine wave.

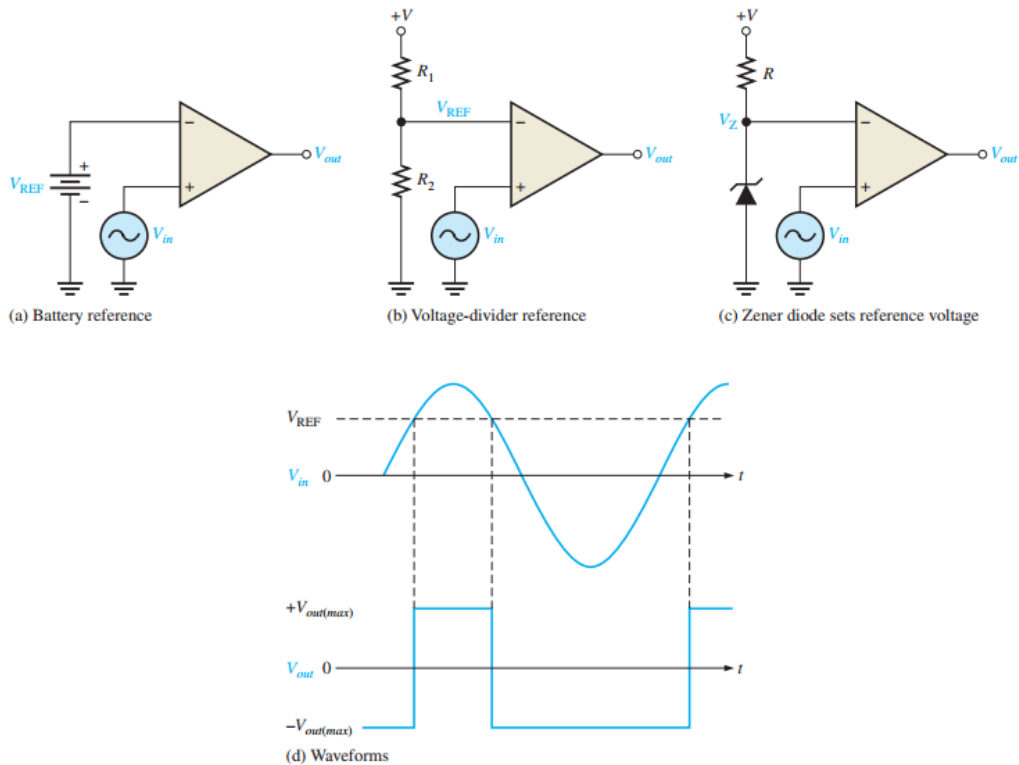


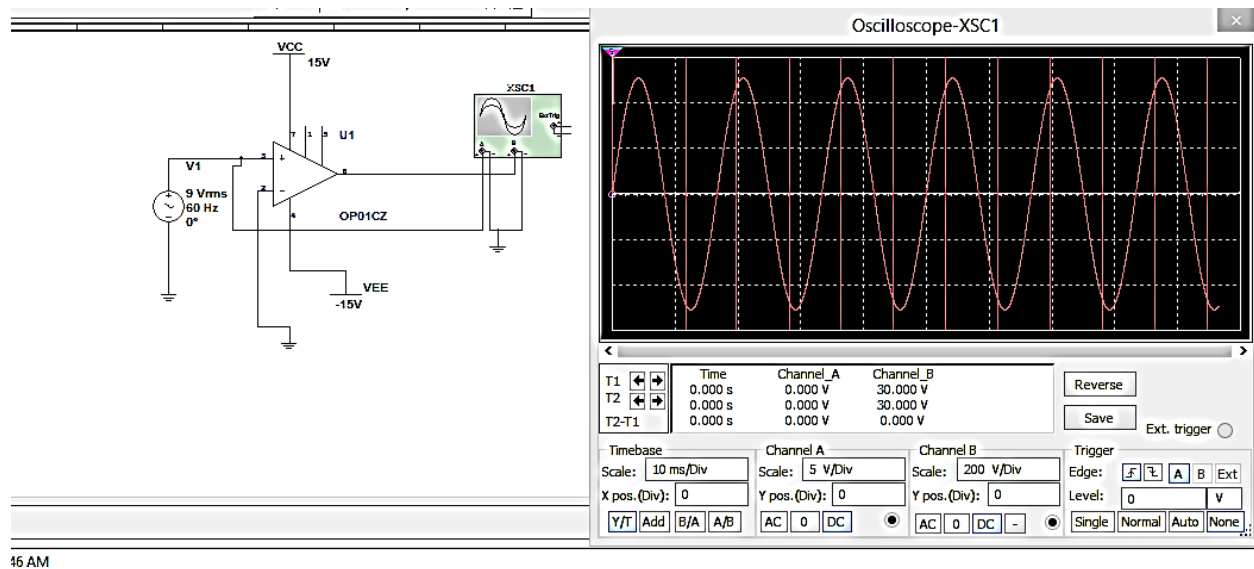
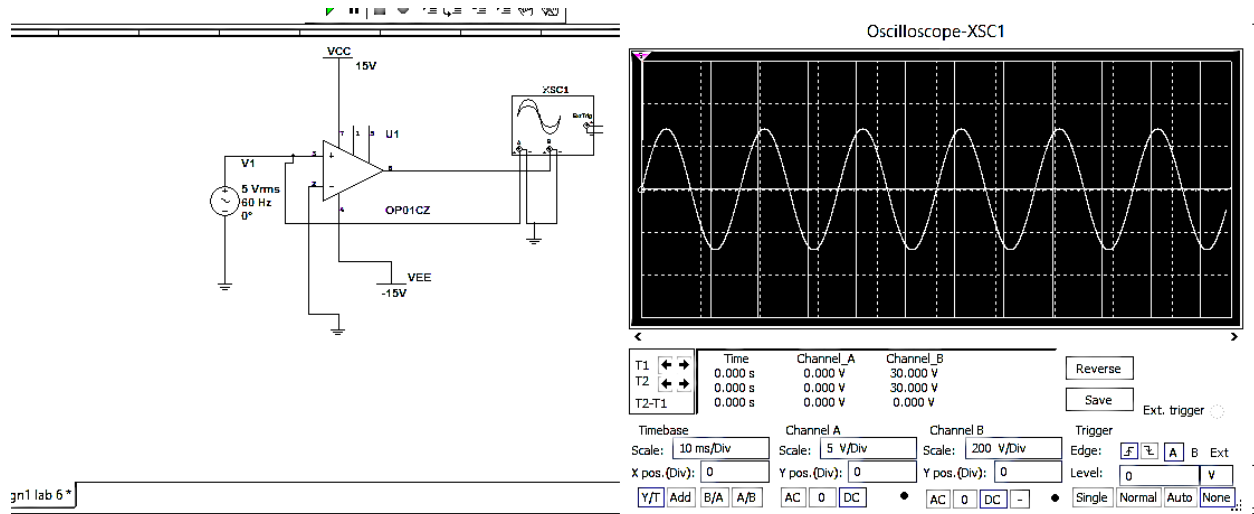
Figure-6.2 Non-Zero level detector with input and output waveform

The circuit in Figure 5.2(c) uses a zener diode to set the reference voltage ($V_{REF} = V_Z$). As long as the input voltage V_{IN} is less than V_{REF} the output remains at the maximum negative level. When the input voltage exceeds the reference voltage, the output goes to its maximum positive state with a sinusoidal input voltage.

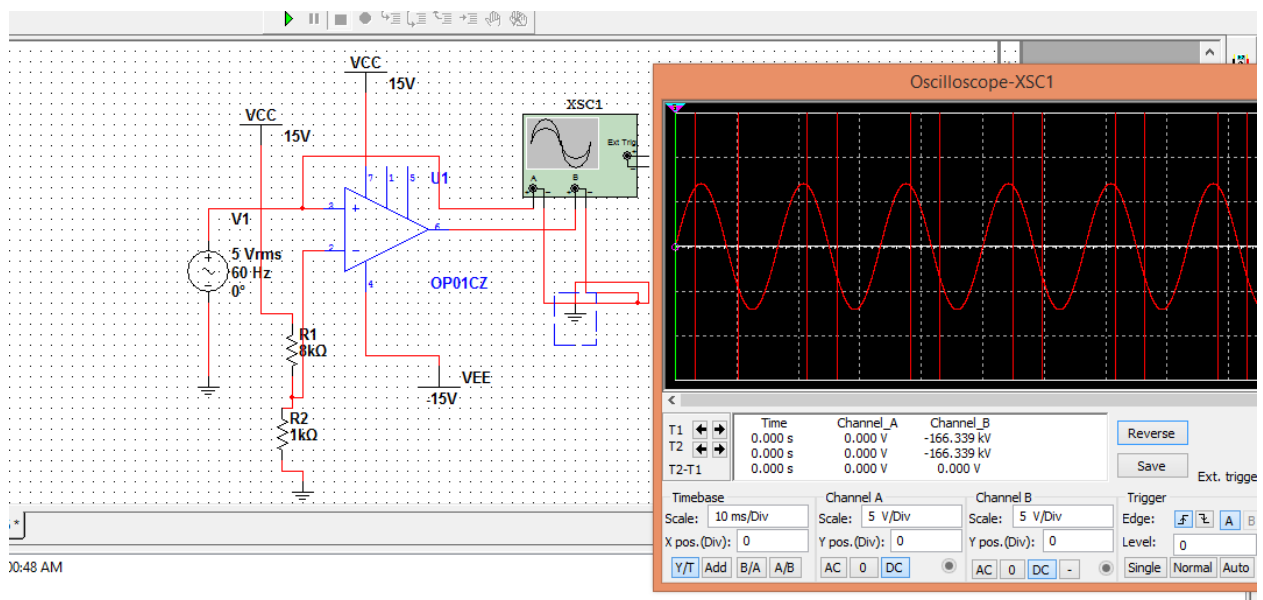
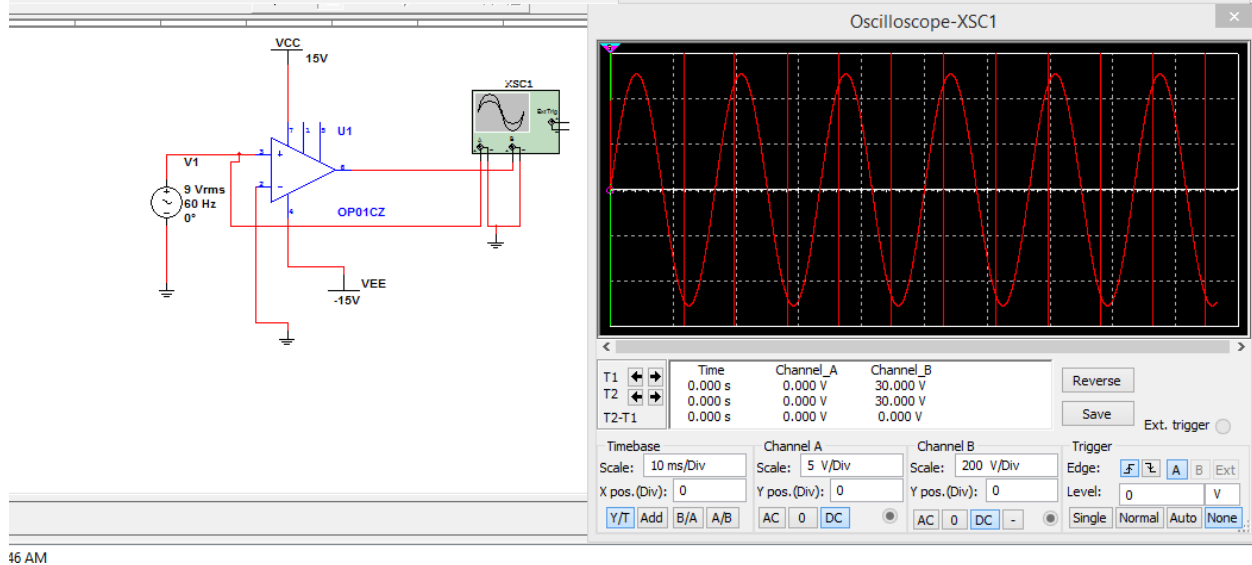
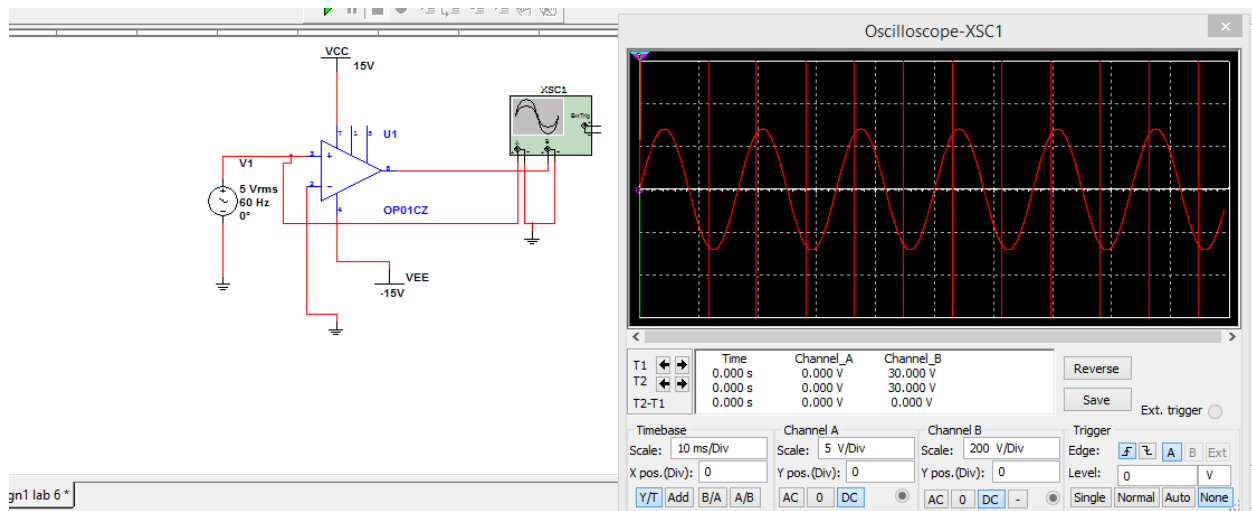
PROCEDURE:

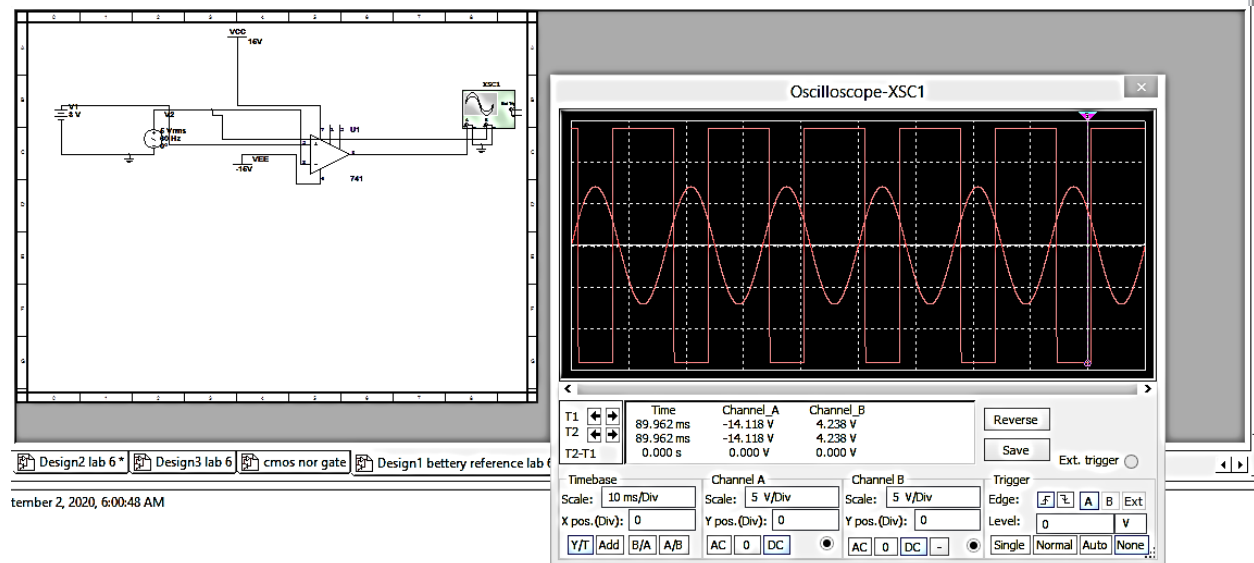
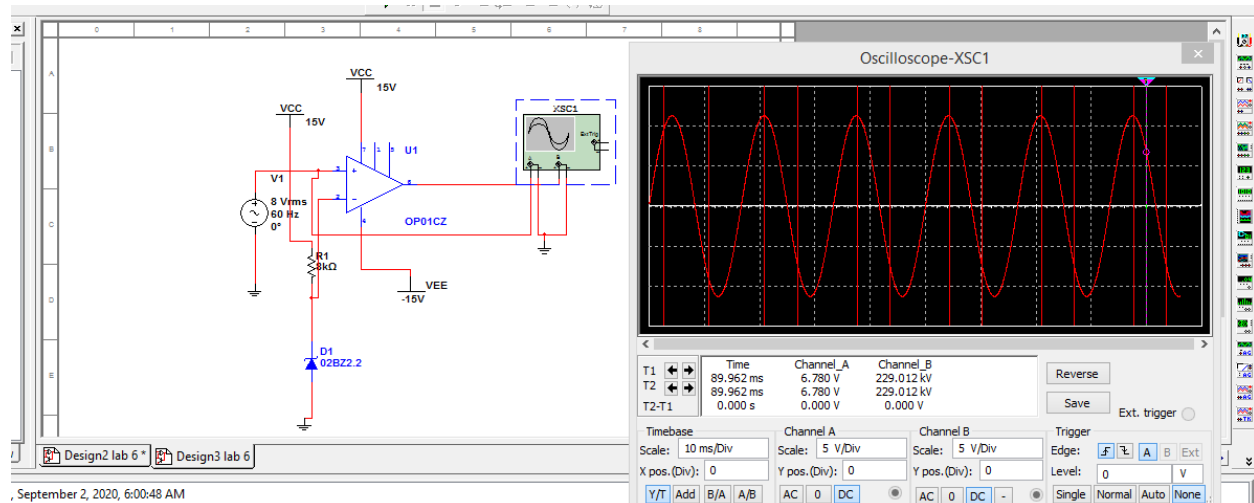
1. Connect the circuit as shown in figure 1.
2. Apply Input voltages at inputs.
3. By changing input values observe the output on oscilloscope.
4. Repeat the same procedure for figure 2.

Lab Activity #1 Take the snapshots of the input and output waveform for the tasks mentioned in 3 and 4 (given in procedure) and attach them with this handout.



Lab Activity #2 Simulate the circuits in above figure-using NI Multisim and attach the results with this handout.

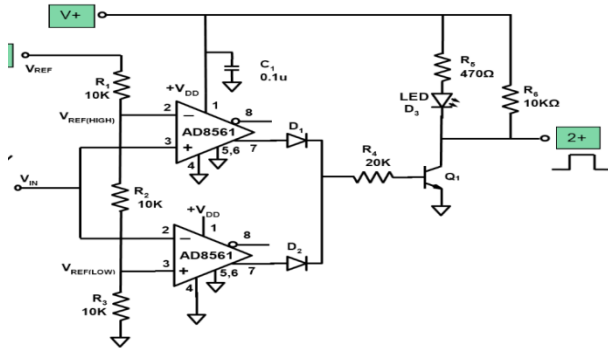




1. What outcomes you achieved from this experiment?

ANS :this experiment is that, if we give some reference voltage to the circuit, if reference voltage is greater than input we give negative output, if input voltage greater than reference then output will be positive. We can change reference voltage by two method using voltage divider bias ,different application can be designed as thermo switches etc using these op-amp application..

Design an application of this experiment on your own choice.



ANS: The circuit uses a voltage divider network, formed of three equal value resistors $R_1 = R_2 = R_3$. The voltage drops across each resistor will also be equal at one-third of the reference voltage (V_{REF}). Therefore, the upper reference ($V_{REF(HIGH)}$) is set to $2/3V_{REF}$ and the lower reference to $1/3V_{REF}$.

Considering that we use the When V_{IN} is below the lower voltage level, ($V_{REF(LOW)}$) which equates to $1/3V_{REF}$, the output will be HIGH and D_2 will be forward biased. Due to the positive voltage at base the NPN transistor, Q_1 moves into the saturation. Thus, the output voltage is zero, and the supply voltage will drop on R_5 and D_3 , turning the LED on.

When V_{IN} exceeds this $1/3V_{REF}$ lower voltage level and it is below $2/3V_{REF}$ ($V_{REF(HIGH)}$), both comparators' outputs will be LOW and the diodes reverse-biased. No voltage is applied to the base of Q_1 , the transistor is in cut-off and no collector current flows through R_6 or R_5 , D_3 . The output voltage is the supply voltage $V+$.

When V_{IN} is above the upper voltage level, ($V_{REF(HIGH)}$) which equates to $2/3V_{REF}$, the output will be HIGH and D_1 will be forward biased. Due to the positive voltage at base the NPN transistor, Q_1 moves into the saturation. Thus, the output voltage is zero, and the supply voltage will drop on R_5 and D_3 , turning the LED on.